wavefront of monochromatic light with wavelength λ be incident on the grating. As the width of the slit is comparable to that of wavelength, the incident light undergoes diffraction.

A diffraction pattern is obtained on the screen when the diffracted waves are focused on a screen using a convex lens. Let us consider a point P at an angle θ with the perpendicular drawn from the center of the grating to the screen. The path difference δ between the diffracted waves from one pair of adjacent corresponding points is,

$$\delta = (a + b) \sin\theta \qquad (7.46)$$

This path difference is the same for any pair of adjacent corresponding points. The point P on the screen will be maximum when,

$$\delta = m\,\lambda \text{ where } m = 0, 1, 2, 3 \qquad (7.47)$$

Combining the above two equations, we get,

$$(a + b) \sin\theta = m\,\lambda \qquad (7.48)$$

Here, m is called order of diffraction maximum.

Condition for P to be zeroth maximum, m = 0

Now, $(a+b)\ \sin\theta = 0$ thus, $\sin\theta = 0$, its position, $\theta = 0$. This is called zeroth diffraction (or) central maximum. It is formed at an angle 0.

Condition for P to be first maximum, m = 1

Now, $(a+b)\ \sin\theta_1 = \lambda$. The first maximum is obtained at an angle θ_1.

Condition for P to be second maximum, m = 2

Now, $(a+b)\ \sin\theta_2 = 2\lambda$. The second maximum is obtained at an angle θ_2.

Condition for P to be mth maximum

On either side of central maximum, different higher order diffraction maxima are formed at different angular positions.

If we take,

$$N = \frac{1}{a+b} \qquad (7.49)$$

Then, N gives the number of grating elements or rulings drawn per unit width of the grating. Normally, this number N is specified on the grating itself. Now, the equation becomes,

$$\frac{1}{N}\sin\theta = m\lambda \text{ (or) } \sin\theta = N\,m\,\lambda \qquad (7.50)$$

Note

The students should remember that in a single slit experiment, the formula, $a\ \sin\theta = n\lambda$ is condition for minimum with n as order of minimum. But in the grating experiment, the formula, $\sin\theta = Nm\lambda$ is condition for maximum with m as the order of diffraction.

EXAMPLE 7.12

A diffraction grating consists of 4000 slits per centimeter. It is illuminated by a monochromatic light. The second order diffraction maximum is produced at an angle of 30°. What is the wavelength of the light used?

Solution

Number of lines = 4000 cm^{-1}; $m = 2$;

$\theta = 30°$; $\lambda = ?$

Number of lines per unit length,

$$N = \frac{4000}{1\times10^{-2}} = 4\times10^{5}\ \text{m}^{-1}$$

Equation for diffraction maximum for grating is, $\sin\theta = Nm\lambda$

After rewriting, $\lambda = \dfrac{\sin\theta}{Nm}$

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